Chapter 2. Linear Transformations and Matrices

Every linear map is a matrix once bases are chosen

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2.1 Linear transformations, null spaces, and ranges

Motivation. Maps preserving addition and scalar multiplication are the morphisms of vector spaces. Their null space and range are subspaces, their dimensions add up to $\dim V$, and a linear map is pinned down by its values on a basis.

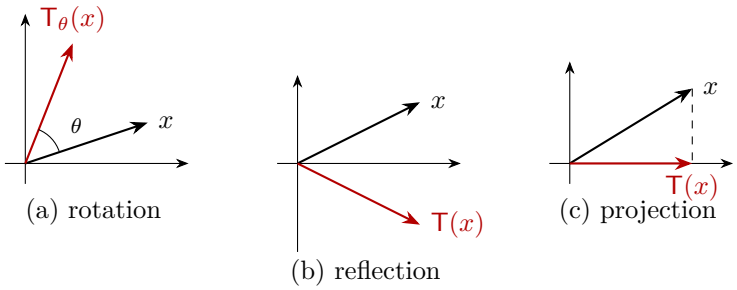
Definition (Linear transformation)

$\mathsf{T} : V \to W$ is **linear** if $\mathsf{T}(x+y) = \mathsf{T}(x) + \mathsf{T}(y)$ and $\mathsf{T}(cx) = c\mathsf{T}(x)$ for all $x, y \in V$, $c \in \mathbb{F}$.

For linear T: $\mathsf{T}(0) = 0$; $\mathsf{T}(cx+y) = c\mathsf{T}(x) + \mathsf{T}(y)$ (and this characterises linearity); $\mathsf{T}(x-y) = \mathsf{T}(x) - \mathsf{T}(y)$; $\mathsf{T}(\sum a_i x_i) = \sum a_i \mathsf{T}(x_i)$.

Example 1

$\mathsf{T}(a_1, a_2) = (2a_1 + a_2, a_1)$ is linear: with $c \in \mathbb{R}$, $x = (b_1, b_2)$, $y = (d_1, d_2)$, $\mathsf{T}(cx+y) = (2(cb_1 + d_1) + cb_2 + d_2,\ cb_1 + d_1) = c\mathsf{T}(x) + \mathsf{T}(y)$.



(a) rotation (b) reflection (c) projection

Figure. *Examples 2–4: rotation by θ, reflection about the x-axis, projection on the x-axis.*

Example 2

The rotation T_θ by angle θ is $\mathsf{T}_\theta(a_1, a_2) = (a_1 \cos\theta - a_2 \sin\theta,\ a_1 \sin\theta + a_2 \cos\theta)$ (write $(a_1, a_2) = r(\cos\alpha, \sin\alpha)$, then the image is $r(\cos(\alpha+\theta), \sin(\alpha+\theta))$). It is linear.

Example 3

The **reflection about the x-axis** $\mathsf{T}(a_1, a_2) = (a_1, -a_2)$ is linear.

Example 4

The **projection on the x-axis** $\mathsf{T}(a_1, a_2) = (a_1, 0)$ is linear.

Example 5

$\mathsf{T} : \mathrm{M}_{m\times n}(\mathbb{F}) \to \mathrm{M}_{n\times m}(\mathbb{F})$, $\mathsf{T}(A) = A^t$, is linear.

Example 6

Differentiation $\mathsf{T}(f) = f'$ from $\mathsf{P}_n(\mathbb{R})$ to $\mathsf{P}_{n-1}(\mathbb{R})$ is linear: $(ag+h)' = ag' + h'$.

Example 7

$\mathsf{T}(f) = \int_a^b f(t)\,dt$ from $C(\mathbb{R})$ to $\mathbb{R}$ is linear.

The **identity** $\mathsf{I}_V(x) = x$ and the **zero transformation** $\mathsf{T}_0(x) = 0$ are linear.

Definition (Null space, range)

For linear $\mathsf{T} : V \to W$, the **null space** (kernel) is $\mathrm{N}(\mathsf{T}) = \{x \in V : \mathsf{T}(x) = 0\}$ and the **range** (image) is $\mathrm{R}(\mathsf{T}) = \{\mathsf{T}(x) : x \in V\}$.

Example 8

$\mathrm{N}(\mathsf{I}) = \{0\}$, $\mathrm{R}(\mathsf{I}) = V$; $\mathrm{N}(\mathsf{T}_0) = V$, $\mathrm{R}(\mathsf{T}_0) = \{0\}$.

Example 9

For $\mathsf{T}(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ on $\mathbb{R}^3$: $\mathrm{N}(\mathsf{T}) = \{(a, a, 0) : a \in \mathbb{R}\}$ and $\mathrm{R}(\mathsf{T}) = \mathbb{R}^2$.

Theorem 2.1

For linear $\mathsf{T} : V \to W$, $\mathrm{N}(\mathsf{T})$ and $\mathrm{R}(\mathsf{T})$ are subspaces of V and W.

Proof. $\mathsf{T}(0) = 0$, so 0 lies in both. If $x, y \in \mathrm{N}(\mathsf{T})$, $c \in \mathbb{F}$, then $\mathsf{T}(x + y) = 0$ and $\mathsf{T}(cx) = 0$. If $\mathsf{T}(x), \mathsf{T}(y) \in \mathrm{R}(\mathsf{T})$, then $\mathsf{T}(x)+\mathsf{T}(y) = \mathsf{T}(x+y)$ and $c\mathsf{T}(x) = \mathsf{T}(cx)$ lie in $\mathrm{R}(\mathsf{T})$. Apply Theorem 1.3. ■

Theorem 2.2

If $\beta = \{v_1, \ldots, v_n\}$ is a basis for V and $\mathsf{T} : V \to W$ is linear, then $\mathrm{R}(\mathsf{T}) = \mathrm{span}(\mathsf{T}(\beta)) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\}$.

Proof. $\mathsf{T}(v_i) \in \mathrm{R}(\mathsf{T})$, a subspace, so $\mathrm{span}(\mathsf{T}(\beta)) \subseteq \mathrm{R}(\mathsf{T})$ by Theorem 1.5. Conversely $w = \mathsf{T}(v)$ with $v = \sum a_i v_i$ gives $w = \sum a_i \mathsf{T}(v_i)$. ■

(The same holds for an infinite basis.)

Example 10

For $\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathrm{M}_{2\times 2}(\mathbb{R})$, $\mathsf{T}(f) = \begin{pmatrix} f(1)-f(2) & 0 \\ 0 & f(0) \end{pmatrix}$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(1), \mathsf{T}(x), \mathsf{T}(x^2)\} = \mathrm{span}\left\{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} -3 & 0 \\ 0 & 0 \end{pmatrix}\right\}$, of dimension 2.

Definition (Nullity, rank)

If $\mathrm{N}(\mathsf{T})$, $\mathrm{R}(\mathsf{T})$ are finite-dimensional, $\mathrm{nullity}(\mathsf{T}) = \dim \mathrm{N}(\mathsf{T})$ and $\mathrm{rank}(\mathsf{T}) = \dim \mathrm{R}(\mathsf{T})$.

Theorem 2.3 (Dimension theorem)

If $\mathsf{T} : V \to W$ is linear and V is finite-dimensional, then $\mathrm{nullity}(\mathsf{T}) + \mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. Let $\dim V = n$ and let $\{v_1, \dots, v_k\}$ be a basis of $\mathrm{N}(\mathsf{T})$; extend it to a basis $\beta = \{v_1, \dots, v_n\}$ of V (Corollary). We claim $S = \{\mathsf{T}(v_{k+1}), \dots, \mathsf{T}(v_n)\}$ is a basis of $\mathrm{R}(\mathsf{T})$ with $n-k$ distinct elements.

Step 1 (spans). By Theorem 2.2, $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(v_1), \dots, \mathsf{T}(v_n)\} = \mathrm{span}(S)$, as $\mathsf{T}(v_i) = 0$ for $i \le k$.

Step 2 (independent). If $\sum_{i>k} b_i \mathsf{T}(v_i) = 0$, then $\sum_{i>k} b_i v_i \in \mathrm{N}(\mathsf{T})$, so $\sum_{i>k} b_i v_i = \sum_{i \le k} c_i v_i$. Independence of β gives all $b_i = 0$ (and shows the $\mathsf{T}(v_i)$, $i > k$, are distinct). ■

Theorem 2.4

Linear T is one-to-one iff $\mathrm{N}(\mathsf{T}) = \{0\}$.

Proof. If T is one-to-one and $\mathsf{T}(x) = 0 = \mathsf{T}(0)$, then $x = 0$. Conversely, if $\mathrm{N}(\mathsf{T}) = \{0\}$ and $\mathsf{T}(x) = \mathsf{T}(y)$, then $\mathsf{T}(x - y) = 0$, so $x = y$. ■

Theorem 2.5

If $\dim V = \dim W < \infty$ and $\mathsf{T} : V \to W$ is linear, the following are equivalent: (a) T is one-to-one;
(b) T is onto;
(c) $\mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. By Theorems 2.3 and 2.4: T one-to-one $\iff$ $\mathrm{N}(\mathsf{T}) = \{0\}$ $\iff$ $\mathrm{nullity}(\mathsf{T}) = 0$ $\iff$ $\mathrm{rank}(\mathsf{T}) = \dim V = \dim W$ $\iff$ $\mathrm{R}(\mathsf{T}) = W$, the last step by Theorem 1.11. ■

Both finite-dimensionality and linearity are essential here.

Example 11

$\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathsf{P}_3(\mathbb{R})$, $\mathsf{T}(f) = 2f' + \int_0^x 3f(t)\,dt$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{3x,\ 2 + \frac{3}{2}x^2,\ 4x + x^3\}$ has rank $3 < 4$, so T is not onto; nullity $= 3 - 3 = 0$, so T is one-to-one.

Example 12

$\mathsf{T}(a_1, a_2) = (a_1 + a_2, a_1)$ on $\mathbb{F}^2$ has $\mathrm{N}(\mathsf{T}) = \{0\}$; by Theorem 2.5 it is also onto.

A one-to-one linear T maps S to an independent set iff S is independent.

Example 13

$\mathsf{T}(a_0 + a_1 x + a_2 x^2) = (a_0, a_1, a_2)$ is linear and one-to-one. $S = \{2 - x + 3x^2,\ x + x^2,\ 1 - 2x^2\}$ is independent since $\mathsf{T}(S) = \{(2, -1, 3), (0, 1, 1), (1, 0, -2)\}$ is.

Theorem 2.6

Let $\{v_1, \dots, v_n\}$ be a basis for V. For any $w_1, \dots, w_n \in W$ there is exactly one linear $\mathsf{T} : V \to W$ with $\mathsf{T}(v_i) = w_i$.

Proof. For $x = \sum a_i v_i$ (unique coefficients, Theorem 1.8) set $\mathsf{T}(x) = \sum a_i w_i$.

Step 1 (linear). If $y = \sum b_i v_i$, then $cx + y = \sum (ca_i + b_i) v_i$, so $\mathsf{T}(cx + y) = \sum (ca_i + b_i) w_i = c\mathsf{T}(x) + \mathsf{T}(y)$.

Step 2 (values). $\mathsf{T}(v_i) = w_i$.

Step 3 (unique). A linear U with $\mathsf{U}(v_i) = w_i$ satisfies $\mathsf{U}(x) = \sum a_i \mathsf{U}(v_i) = \sum a_i w_i = \mathsf{T}(x)$. ■

Corollary

If $\{v_1, \dots, v_n\}$ is a basis for V and linear $\mathsf{U}, \mathsf{T} : V \to W$ agree on each v_i, then $\mathsf{U} = \mathsf{T}$.

Proof. Uniqueness in Theorem 2.6. ■

Example 14

If $\mathsf{U} : \mathbb{R}^2 \to \mathbb{R}^2$ is linear with $\mathsf{U}(1,2) = (3,3)$ and $\mathsf{U}(1,1) = (1,3)$, then $\mathsf{U} = \mathsf{T}$ for $\mathsf{T}(a_1, a_2) = (2a_2 - a_1, 3a_1)$, since T takes the same values on the basis $\{(1,2),(1,1)\}$.

2.2 The matrix representation of a linear transformation

Motivation. Once ordered bases are fixed, coordinates turn vectors into columns and linear maps into matrices: the j-th column of $[\mathsf{T}]_\beta^\gamma$ lists the coordinates of $\mathsf{T}(v_j)$. Sums and scalar multiples of maps correspond to those of matrices.

Definition (Ordered basis; coordinate vector)

An **ordered basis** is a basis with a fixed order. For $\beta = \{u_1, \dots, u_n\}$ and $x = \sum a_i u_i$, the **coordinate vector** is $[x]_\beta = (a_1, \dots, a_n)^t \in \mathbb{F}^n$. The **standard ordered bases** are $\{e_1, \dots, e_n\}$ of $\mathbb{F}^n$ and $\{1, x, \dots, x^n\}$ of $\mathsf{P}_n(\mathbb{F})$.

Example 1

$\beta = \{e_1, e_2, e_3\}$ and $\gamma = \{e_2, e_1, e_3\}$ are different ordered bases of $\mathbb{F}^3$.

Example 2

For $f = 4 + 6x - 7x^2$ and the standard β of $\mathsf{P}_2(\mathbb{R})$, $[f]_\beta = (4, 6, -7)^t$.

Definition (Matrix representation)

Let $\beta = \{v_1, \dots, v_n\}$, $\gamma = \{w_1, \dots, w_m\}$ be ordered bases of V, W and $\mathsf{T} : V \to W$ linear, with $\mathsf{T}(v_j) = \sum_{i=1}^m a_{ij} w_i$. The $m \times n$ matrix $A = (a_{ij}) = [\mathsf{T}]_\beta^\gamma$ is the **matrix representation** of T; its j-th column is $[\mathsf{T}(v_j)]_\gamma$. If $V = W$, $\beta = \gamma$, write $[\mathsf{T}]_\beta$.

By the Corollary to Theorem 2.6, $[\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma$ implies $\mathsf{U} = \mathsf{T}$.

Example 3

$\mathsf{T}(a_1, a_2) = (a_1 + 3a_2, 0, 2a_1 - 4a_2)$ in the standard bases: $\mathsf{T}(e_1) = (1, 0, 2)$, $\mathsf{T}(e_2) = (3, 0, -4)$, so $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}$. With $\gamma' = \{e_3, e_2, e_1\}$ the rows are reversed.

Example 4

$\mathsf{T}(f) = f'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$, standard bases: $\mathsf{T}(1) = 0$, $\mathsf{T}(x) = 1$, $\mathsf{T}(x^2) = 2x$, $\mathsf{T}(x^3) = 3x^2$, so $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

Definition (Sum and scalar multiple of maps)

$(\mathsf{T}+\mathsf{U})(x)=\mathsf{T}(x)+\mathsf{U}(x)$ and $(a\mathsf{T})(x)=a\mathsf{T}(x)$.

Theorem 2.7

Let $\mathsf{T},\mathsf{U}:V\to W$ be linear.
(a) $a\mathsf{T}+\mathsf{U}$ is linear for all $a\in\mathbb{F}$.
(b) With these operations, the linear maps $V\to W$ form a vector space over $\mathbb{F}$.

Proof.

(a) $(a\mathsf{T}+\mathsf{U})(cx+y)=a\mathsf{T}(cx+y)+\mathsf{U}(cx+y)=c(a\mathsf{T}+\mathsf{U})(x)+(a\mathsf{T}+\mathsf{U})(y)$.

(b) By (a) the linear maps form a subset of $\mathcal{F}(V,W)$ containing T_0 and closed under the operations; the axioms hold pointwise since they hold in W. ■

$\mathcal{L}(V,W)$ denotes this space, and $\mathcal{L}(V)=\mathcal{L}(V,V)$.

Theorem 2.8

For finite-dimensional V,W with ordered bases β,γ and linear $\mathsf{T},\mathsf{U}:V\to W$:
(a) $[\mathsf{T}+\mathsf{U}]_\beta^\gamma=[\mathsf{T}]_\beta^\gamma+[\mathsf{U}]_\beta^\gamma$;
(b) $[a\mathsf{T}]_\beta^\gamma=a[\mathsf{T}]_\beta^\gamma$.

Proof. If $\mathsf{T}(v_j)=\sum_i a_{ij}w_i$ and $\mathsf{U}(v_j)=\sum_i b_{ij}w_i$, then $(\mathsf{T}+\mathsf{U})(v_j)=\sum_i(a_{ij}+b_{ij})w_i$ and $(a\mathsf{T})(v_j)=\sum_i aa_{ij}w_i$. ■

Example 5

For $\mathsf{T}(a_1,a_2)=(a_1+3a_2,0,2a_1-4a_2)$ and $\mathsf{U}(a_1,a_2)=(a_1-a_2,2a_1,3a_1+2a_2)$, $(\mathsf{T}+\mathsf{U})(a_1,a_2)=(2a_1+2a_2,2a_1,5a_1-2a_2)$, and in the standard bases its matrix $\begin{pmatrix}2&2\\2&0\\5&-2\end{pmatrix}$ is the sum of $\begin{pmatrix}1&3\\0&0\\2&-4\end{pmatrix}$ and $\begin{pmatrix}1&-1\\2&0\\3&2\end{pmatrix}$.

2.3 Composition of linear transformations and matrix multiplication

Motivation. Matrix multiplication is defined exactly so that composing maps corresponds to multiplying their matrices. Associativity of matrix products then comes for free from associativity of composition.

Theorem 2.9

If $\mathsf{T}:V\to W$ and $\mathsf{U}:W\to Z$ are linear, so is $\mathsf{UT}:V\to Z$.

Proof. $\mathsf{UT}(ax+y)=\mathsf{U}(a\mathsf{T}(x)+\mathsf{T}(y))=a\mathsf{UT}(x)+\mathsf{UT}(y)$. ■

Theorem 2.10

For $\mathsf{T},\mathsf{U}_1,\mathsf{U}_2\in\mathcal{L}(V)$:
(a) $\mathsf{T}(\mathsf{U}_1+\mathsf{U}_2)=\mathsf{TU}_1+\mathsf{TU}_2$ and $(\mathsf{U}_1+\mathsf{U}_2)\mathsf{T}=\mathsf{U}_1\mathsf{T}+\mathsf{U}_2\mathsf{T}$;

(b) $\mathsf{T}(\mathsf{U}_1\mathsf{U}_2) = (\mathsf{T}\mathsf{U}_1)\mathsf{U}_2$;
(c) $\mathsf{T}\mathsf{I} = \mathsf{I}\mathsf{T} = \mathsf{T}$;
(d) $a(\mathsf{U}_1\mathsf{U}_2) = (a\mathsf{U}_1)\mathsf{U}_2 = \mathsf{U}_1(a\mathsf{U}_2)$.

Proof [Supplement]. Evaluate at x. (a) $\mathsf{T}(\mathsf{U}_1x + \mathsf{U}_2x) = \mathsf{T}\mathsf{U}_1x + \mathsf{T}\mathsf{U}_2x$ by linearity of T; $(\mathsf{U}_1 + \mathsf{U}_2)(\mathsf{T}x) = \mathsf{U}_1\mathsf{T}x + \mathsf{U}_2\mathsf{T}x$ by definition.

(b) Composition of functions is associative.

(c) Clear.

(d) $a\mathsf{U}_1(\mathsf{U}_2x) = (a\mathsf{U}_1)(\mathsf{U}_2x) = \mathsf{U}_1(a\mathsf{U}_2x)$ by linearity of U_1. ■

Let $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$ with ordered bases $\alpha = \{v_j\}$, $\beta = \{w_k\}$, $\gamma = \{z_i\}$, $A = [\mathsf{U}]^\gamma_\beta$, $B = [\mathsf{T}]^\beta_\alpha$. Then

$$\mathsf{U}\mathsf{T}(v_j) = \mathsf{U}\Big(\sum_k B_{kj}w_k\Big) = \sum_k B_{kj}\sum_i A_{ik}z_i = \sum_i \Big(\sum_k A_{ik}B_{kj}\Big)z_i.$$

Definition (Matrix product)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $B \in \mathrm{M}_{n\times p}(\mathbb{F})$, $AB \in \mathrm{M}_{m\times p}(\mathbb{F})$ has $(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj}$.

Example 1

$\begin{pmatrix}1&2&1\\0&4&-1\end{pmatrix}\begin{pmatrix}4\\2\\5\end{pmatrix} = \begin{pmatrix}13\\3\end{pmatrix}$. Matrix multiplication is not commutative: $\begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}1&1\\0&0\end{pmatrix}$ but $\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}0&0\\1&1\end{pmatrix}$. Also $(AB)^t = B^tA^t$, since both have (i,j) entry $\sum_k A_{jk}B_{ki}$.

Theorem 2.11

With α, β, γ ordered bases of finite-dimensional V, W, Z and linear $\mathsf{T} : V \to W$, $\mathsf{U} : W \to Z$: $[\mathsf{U}\mathsf{T}]^\gamma_\alpha = [\mathsf{U}]^\gamma_\beta[\mathsf{T}]^\beta_\alpha$.

Proof. The computation preceding the definition: the (i,j) entry of $[\mathsf{U}\mathsf{T}]^\gamma_\alpha$ is $\sum_k A_{ik}B_{kj}$. ■

Corollary

For $\mathsf{T}, \mathsf{U} \in \mathcal{L}(V)$ and an ordered basis β, $[\mathsf{U}\mathsf{T}]_\beta = [\mathsf{U}]_\beta[\mathsf{T}]_\beta$.

Proof. Theorem 2.11 with $\alpha = \beta = \gamma$. ■

Example 2

$\mathsf{U}(f) = f'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$ and $\mathsf{T}(f) = \int_0^x f(t)\,dt$ from $\mathsf{P}_2(\mathbb{R})$ to $\mathsf{P}_3(\mathbb{R})$ give $\mathsf{U}\mathsf{T} = \mathsf{I}$. Indeed, in standard bases

$$[\mathsf{U}][\mathsf{T}] = \begin{pmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}\begin{pmatrix}0&0&0\\1&0&0\\0&\frac12&0\\0&0&\frac13\end{pmatrix} = I_3.$$

Definition (Kronecker delta; identity matrix)

$\delta_{ij} = 1$ if $i = j$ and 0 otherwise; the $n \times n$ **identity matrix** is $(I_n)_{ij} = \delta_{ij}$.

Theorem 2.12

Let A be $m \times n$, B, C be $n \times p$, D, E be $q \times m$. Then
(a) $A(B+C) = AB + AC$ and $(D+E)A = DA + EA$;
(b) $a(AB) = (aA)B = A(aB)$;
(c) $I_m A = A = AI_n$;
(d) $[\mathsf{I}_V]_\beta = I_n$ for an n-dimensional V with ordered basis β.

Proof.

(a) $(A(B+C))_{ij} = \sum_k A_{ik}(B_{kj} + C_{kj}) = (AB)_{ij} + (AC)_{ij}$; the other identity is similar.

(b) [Supplement] All three have entries $a\sum_k A_{ik}B_{kj}$.

(c) $(I_m A)_{ij} = \sum_k \delta_{ik}A_{kj} = A_{ij}$, and similarly for AI_n.

(d) [Supplement] $\mathsf{I}_V(v_j) = v_j$, whose coordinate vector is e_j. ■

Corollary

$A(\sum_{i=1}^k a_i B_i) = \sum_i a_i AB_i$ and $(\sum_i a_i C_i)A = \sum_i a_i C_i A$.

Proof. Induction on k using Theorem 2.12(a), (b). ■

Powers: $A^1 = A$, $A^k = A^{k-1}A$, $A^0 = I$. Note $A^2 = O$ is possible with $A \neq O$, e.g. $A = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, so there is no cancellation law.

Theorem 2.13

Let A be $m \times n$, B be $n \times p$, and u_j, v_j the j-th columns of AB and B. Then (a) $u_j = Av_j$; (b) $v_j = Be_j$.

Proof.

(a) $(u_j)_i = (AB)_{ij} = \sum_k A_{ik}B_{kj} = (Av_j)_i$.

(b) Apply (a) with $A = B$, $B = I_p$: Be_j is the j-th column of $BI_p = B$. ■

Similarly the i-th row of AB is (row i of A)$\cdot B$.

Theorem 2.14

For linear $\mathsf{T} : V \to W$ with ordered bases β, γ and $u \in V$: $[\mathsf{T}(u)]_\gamma = [\mathsf{T}]_\beta^\gamma [u]_\beta$.

Proof. Let $f : \mathbb{F} \to V$, $f(a) = au$, and $g : \mathbb{F} \to W$, $g(a) = a\mathsf{T}(u)$, with the basis $\alpha = \{1\}$ of $\mathbb{F}$. Then $g = \mathsf{T}f$, and $[f]_\alpha^\beta = [u]_\beta$, $[g]_\alpha^\gamma = [\mathsf{T}(u)]_\gamma$. By Theorem 2.11, $[\mathsf{T}(u)]_\gamma = [\mathsf{T}f]_\alpha^\gamma = [\mathsf{T}]_\beta^\gamma [f]_\alpha^\beta = [\mathsf{T}]_\beta^\gamma [u]_\beta$. ■

Example 3

For $\mathsf{T}(f) = f'$ on $\mathsf{P}_3(\mathbb{R})$ and $p = 2 - 4x + x^2 + 3x^3$: $[\mathsf{T}(p)]_\gamma = [\mathsf{T}]_\beta^\gamma (2, -4, 1, 3)^t = (-4, 2, 9)^t$, matching $p' = -4 + 2x + 9x^2$.

Definition (Left-multiplication transformation)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $\mathsf{L}_A : \mathbb{F}^n \to \mathbb{F}^m$ is $\mathsf{L}_A(x) = Ax$.

Example 4

For $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$, $\mathsf{L}_A(1,3,-1)^t = (6,1)^t$.

Theorem 2.15

L_A is linear. For $B \in \mathrm{M}_{m\times n}(\mathbb{F})$ and standard ordered bases β, γ of $\mathbb{F}^n, \mathbb{F}^m$:
(a) $[\mathsf{L}_A]^\gamma_\beta = A$;
(b) $\mathsf{L}_A = \mathsf{L}_B$ iff $A = B$;
(c) $\mathsf{L}_{A+B} = \mathsf{L}_A + \mathsf{L}_B$ and $\mathsf{L}_{aA} = a\mathsf{L}_A$;
(d) if $\mathsf{T} : \mathbb{F}^n \to \mathbb{F}^m$ is linear, there is a unique C with $\mathsf{T} = \mathsf{L}_C$, namely $C = [\mathsf{T}]^\gamma_\beta$;
(e) if E is $n \times p$, then $\mathsf{L}_{AE} = \mathsf{L}_A\mathsf{L}_E$;
(f) if $m = n$, then $\mathsf{L}_{I_n} = \mathsf{I}_{\mathbb{F}^n}$.

Proof. Linearity follows from Theorem 2.12.

(a) The j-th column of $[\mathsf{L}_A]^\gamma_\beta$ is $\mathsf{L}_A(e_j) = Ae_j$, the j-th column of A (Theorem 2.13).

(b) If $\mathsf{L}_A = \mathsf{L}_B$, then $A = [\mathsf{L}_A]^\gamma_\beta = [\mathsf{L}_B]^\gamma_\beta = B$ by (a).

(c) [Supplement] $(A+B)x = Ax + Bx$ and $(aA)x = a(Ax)$.

(d) Put $C = [\mathsf{T}]^\gamma_\beta$. By Theorem 2.14, $\mathsf{T}(x) = [\mathsf{T}(x)]_\gamma = C[x]_\beta = Cx = \mathsf{L}_C(x)$. Uniqueness is (b).

(e) For each j, $(AE)e_j = A(Ee_j)$ by Theorem 2.13; so L_{AE} and $\mathsf{L}_A\mathsf{L}_E$ agree on a basis, and are equal by the Corollary to Theorem 2.6.

(f) [Supplement] $I_n x = x$. ■

Theorem 2.16

If $A(BC)$ is defined, so is $(AB)C$, and $A(BC) = (AB)C$.

Proof. By Theorem 2.15(e) and associativity of composition, $\mathsf{L}_{A(BC)} = \mathsf{L}_A\mathsf{L}_{BC} = \mathsf{L}_A(\mathsf{L}_B\mathsf{L}_C) = (\mathsf{L}_A\mathsf{L}_B)\mathsf{L}_C = \mathsf{L}_{(AB)C}$; apply (b). ■

Incidence matrices. For a relation on $\{1,\dots,n\}$ (e.g. "i can send a message to j"), the **incidence matrix** has $A_{ij} = 1$ if i relates to j and 0 otherwise. Then $(A^2)_{ij}$ counts the two-stage paths $i \to k \to j$, and $(A^k)_{ij}$ counts paths of k steps. In a **dominance relation** (for $i \neq j$ exactly one of $i \to j$, $j \to i$; $A_{ii} = 0$), the row sums of $A + A^2$ measure the power of each member.

2.4 Invertibility and isomorphisms

Motivation. An invertible linear map is an isomorphism: it identifies two spaces completely. Finite-dimensional spaces over $\mathbb{F}$ are classified by a single number, the dimension, and $\mathcal{L}(V,W) \cong \mathrm{M}_{m\times n}(\mathbb{F})$.

Definition (Inverse)

$\mathsf{U} : W \to V$ is an **inverse** of linear $\mathsf{T} : V \to W$ if $\mathsf{TU} = \mathsf{I}_W$ and $\mathsf{UT} = \mathsf{I}_V$; T is then **invertible**, the inverse is unique and denoted T^{-1}.

For invertible functions: $(\mathsf{TU})^{-1} = \mathsf{U}^{-1}\mathsf{T}^{-1}$; $(\mathsf{T}^{-1})^{-1} = \mathsf{T}$; and a function is invertible iff it is one-to-one and onto. So if $\dim V = \dim W < \infty$, linear T is invertible iff $\operatorname{rank}(\mathsf{T}) = \dim V$ (Theorem 2.5).

Example 1

$\mathsf{T}(a+bx) = (a, a+b)$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$ has $\mathsf{T}^{-1}(c,d) = c + (d-c)x$.

Theorem 2.17

If $\mathsf{T}: V \to W$ is linear and invertible, then T^{-1} is linear.

Proof. For $y_1, y_2 \in W$ let $x_i = \mathsf{T}^{-1}(y_i)$. Then $\mathsf{T}^{-1}(cy_1+y_2) = \mathsf{T}^{-1}(c\mathsf{T}(x_1)+\mathsf{T}(x_2)) = \mathsf{T}^{-1}(\mathsf{T}(cx_1+x_2)) = cx_1+x_2 = c\mathsf{T}^{-1}(y_1)+\mathsf{T}^{-1}(y_2)$. ■

Definition (Invertible matrix)

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is **invertible** if $AB = BA = I$ for some B. Then B is unique ($C = CI = C(AB) = (CA)B = B$), the **inverse** A^{-1}.

Example 2

$\left(\begin{smallmatrix}5 & 7\\ 2 & 3\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}3 & -7\\ -2 & 5\end{smallmatrix}\right)$.

Lemma

Let $\mathsf{T}: V \to W$ be an invertible linear map. Then V is finite-dimensional iff W is, and then $\dim V = \dim W$.

Proof. If V has a basis $\beta = \{x_1, \dots, x_n\}$, then $\operatorname{span}\mathsf{T}(\beta) = \mathrm{R}(\mathsf{T}) = W$ (Theorem 2.2), so W is finite-dimensional; the converse uses T^{-1}. Then $\operatorname{nullity}\mathsf{T} = 0$ and $\operatorname{rank}\mathsf{T} = \dim W$, so $\dim V = \dim W$ by Theorem 2.3. ■

Theorem 2.18

Let V, W be finite-dimensional with ordered bases β, γ and $\mathsf{T}: V \to W$ linear. Then T is invertible iff $[\mathsf{T}]_\beta^\gamma$ is invertible, and then $[\mathsf{T}^{-1}]_\gamma^\beta = ([\mathsf{T}]_\beta^\gamma)^{-1}$.

Proof.

Step 1 (only if). By the Lemma $\dim V = \dim W = n$, so $[\mathsf{T}]_\beta^\gamma$ is $n \times n$. By Theorems 2.11 and 2.12(d), $I_n = [\mathsf{I}_V]_\beta = [\mathsf{T}^{-1}\mathsf{T}]_\beta = [\mathsf{T}^{-1}]_\gamma^\beta[\mathsf{T}]_\beta^\gamma$, and likewise $[\mathsf{T}]_\beta^\gamma[\mathsf{T}^{-1}]_\gamma^\beta = I_n$.

Step 2 (if). Let $A = [\mathsf{T}]_\beta^\gamma$ have inverse B; then $n = \dim V = \dim W$ (A square). By Theorem 2.6 there is a linear $\mathsf{U}: W \to V$ with $\mathsf{U}(w_j) = \sum_i B_{ij}v_i$, i.e. $[\mathsf{U}]_\gamma^\beta = B$. Then $[\mathsf{UT}]_\beta = BA = I_n = [\mathsf{I}_V]_\beta$, so $\mathsf{UT} = \mathsf{I}_V$, and likewise $\mathsf{TU} = \mathsf{I}_W$. ■

Example 3

For T of Example 1 in standard bases, $[\mathsf{T}] = \left(\begin{smallmatrix}1 & 0\\ 1 & 1\end{smallmatrix}\right)$ and $[\mathsf{T}^{-1}] = \left(\begin{smallmatrix}1 & 0\\ -1 & 1\end{smallmatrix}\right)$, which are inverse to each other.

Corollary 1

For $\mathsf{T} \in \mathcal{L}(V)$, V finite-dimensional with ordered basis β: T is invertible iff $[\mathsf{T}]_\beta$ is, and $[\mathsf{T}^{-1}]_\beta = ([\mathsf{T}]_\beta)^{-1}$.

Proof. Theorem 2.18 with $W = V$, $\gamma = \beta$. ■

Corollary 2

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is invertible iff L_A is, and $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$.

Proof. Theorem 2.18 with standard bases and Theorem 2.15(a), (d): $[(\mathsf{L}_A)^{-1}] = A^{-1}$, so $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$. ■

Definition (Isomorphism)

V is **isomorphic** to W ($V \cong W$) if there is an invertible linear $\mathsf{T} : V \to W$, an **isomorphism**. "Is isomorphic to" is an equivalence relation.

Example 4

$\mathsf{T}(a_1, a_2) = a_1 + a_2 x$ is an isomorphism $\mathbb{F}^2 \to \mathsf{P}_1(\mathbb{F})$.

Example 5

$\mathsf{T}(f) = \begin{pmatrix} f(1) & f(2) \\ f(3) & f(4) \end{pmatrix}$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathrm{M}_{2\times 2}(\mathbb{R})$ is linear, and $\mathsf{T}(f) = O$ only for $f = 0$ (a cubic with four zeros is zero; Lagrange). Since the dimensions agree (4), T is an isomorphism by Theorem 2.5.

Theorem 2.19

Finite-dimensional V, W over $\mathbb{F}$ are isomorphic iff $\dim V = \dim W$.

Proof. "Only if" is the Lemma. If $\dim V = \dim W = n$ with bases $\{v_i\}$, $\{w_i\}$, Theorem 2.6 gives linear T with $\mathsf{T}(v_i) = w_i$. Then $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{w_i\} = W$, so T is onto, hence one-to-one by Theorem 2.5: an isomorphism. ■

Corollary

$V \cong \mathbb{F}^n$ iff $\dim V = n$.

Proof. $\dim \mathbb{F}^n = n$; apply Theorem 2.19 (if $V \cong \mathbb{F}^n$, V is finite-dimensional by the Lemma). ■

Theorem 2.20

Let $\dim V = n$, $\dim W = m$ with ordered bases β, γ. Then $\Phi : \mathcal{L}(V, W) \to \mathrm{M}_{m\times n}(\mathbb{F})$, $\Phi(\mathsf{T}) = [\mathsf{T}]_\beta^\gamma$, is an isomorphism.

Proof. Φ is linear by Theorem 2.8. Given A, Theorem 2.6 gives a unique linear T with $\mathsf{T}(v_j) = \sum_i A_{ij} w_i$, i.e. a unique T with $\Phi(\mathsf{T}) = A$. So Φ is bijective. ■

Corollary

$\dim \mathcal{L}(V, W) = mn$.

Proof. Theorems 2.20 and 2.19. ■

Definition (Standard representation)

For an ordered basis β of n-dimensional V, $\phi_\beta : V \to \mathbb{F}^n$, $\phi_\beta(x) = [x]_\beta$, is the **standard representation** of V with respect to β.

Example 6

For $\beta = \{(1,0),(0,1)\}$, $\gamma = \{(1,2),(3,4)\}$ and $x = (1,-2)$: $\phi_\beta(x) = (1,-2)^t$ and $\phi_\gamma(x) = (-5,2)^t$, since $(1,-2) = -5(1,2) + 2(3,4)$.

Theorem 2.21

ϕ_β is an isomorphism.

Proof [Supplement]. ϕ_β is linear (coordinates of $cx + y$ are $c[x]_\beta + [y]_\beta$ by uniqueness in Theorem 1.8) and $\phi_\beta(x) = 0$ only for $x = 0$. As $\dim V = \dim \mathbb{F}^n$, Theorem 2.5 applies. ■

$$\begin{array}{ccc} V & \xrightarrow{\mathsf{T}} & W \\ {\scriptstyle\phi_\beta}\downarrow & & \downarrow{\scriptstyle\phi_\gamma} \\ \mathbb{F}^n & \xrightarrow[\mathsf{L}_A]{} & \mathbb{F}^m \end{array}$$

Figure. *With $A = [\mathsf{T}]_\beta^\gamma$, the square commutes: $\mathsf{L}_A\phi_\beta = \phi_\gamma\mathsf{T}$ (Theorem 2.14). Computations in V, W become matrix computations.*

Example 7

For $\mathsf{T}(p) = p'$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathsf{P}_2(\mathbb{R})$ and $p = 2 + x - 3x^2 + 5x^3$: $\mathsf{L}_A\phi_\beta(p) = A(2,1,-3,5)^t = (1,-6,15)^t = \phi_\gamma(p')$.

2.5 The change of coordinate matrix

Motivation. Changing the basis changes coordinates by an invertible matrix Q and the matrix of an operator by conjugation $Q^{-1}[\mathsf{T}]Q$. Choosing a clever basis can make $[\mathsf{T}]$ simple; this is the theme of Chapter 5.

Theorem 2.22

Let β, β' be ordered bases of finite-dimensional V and $Q = [\mathsf{I}_V]_{\beta'}^{\beta}$. Then (a) Q is invertible; (b) $[v]_\beta = Q[v]_{\beta'}$ for all $v \in V$.

Proof.

(a) I_V is invertible, so Q is by Theorem 2.18.

(b) $[v]_\beta = [\mathsf{I}_V(v)]_\beta = [\mathsf{I}_V]_{\beta'}^{\beta}[v]_{\beta'}$ by Theorem 2.14. ■

Q is the **change of coordinate matrix** (from β'- to β-coordinates); its j-th column is $[x'_j]_\beta$, and Q^{-1} changes β- into β'-coordinates.

Example 1

In $\mathbb{R}^2$ with $\beta = \{(1,1),(1,-1)\}$, $\beta' = \{(2,4),(3,1)\}$: $(2,4) = 3(1,1) - (1,-1)$ and $(3,1) = 2(1,1) + (1,-1)$, so $Q = \left(\begin{smallmatrix} 3 & 2 \\ -1 & 1 \end{smallmatrix}\right)$.

A linear map $\mathsf{T} : V \to V$ is a **linear operator**.

Theorem 2.23

Let T be a linear operator on finite-dimensional V with ordered bases β, β', and $Q = [\mathsf{I}_V]_{\beta'}^{\beta}$. Then $[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_{\beta} Q$.

Proof. $\mathsf{T} = \mathsf{I}\mathsf{T} = \mathsf{T}\mathsf{I}$, so by Theorem 2.11, $Q[\mathsf{T}]_{\beta'} = [\mathsf{I}]_{\beta'}^{\beta}[\mathsf{T}]_{\beta'}^{\beta'} = [\mathsf{T}]_{\beta'}^{\beta} = [\mathsf{T}]_{\beta}^{\beta}[\mathsf{I}]_{\beta'}^{\beta} = [\mathsf{T}]_{\beta} Q$. Multiply by Q^{-1}. ■

Example 2

For $\mathsf{T}(a,b) = (3a - b, a + 3b)$ and β, β' of Example 1: $[\mathsf{T}]_{\beta} = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 3 \end{smallmatrix}\right)$, $Q^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 1 & -2 \\ 1 & 3 \end{smallmatrix}\right)$, and

$$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_{\beta} Q = \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix}.$$

Check: $\mathsf{T}(2,4) = (2,14) = 4(2,4) - 2(3,1)$.

Example 3

The reflection T about the line $y = 2x$: with $\beta' = \{(1,2),(-2,1)\}$, $\mathsf{T}(1,2) = (1,2)$ and $\mathsf{T}(-2,1) = -(-2,1)$, so $[\mathsf{T}]_{\beta'} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. With β standard, $Q = \left(\begin{smallmatrix} 1 & -2 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$[\mathsf{T}]_{\beta} = Q[\mathsf{T}]_{\beta'} Q^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \mathsf{T}(a,b) = \tfrac{1}{5}(-3a + 4b,\ 4a + 3b).$$

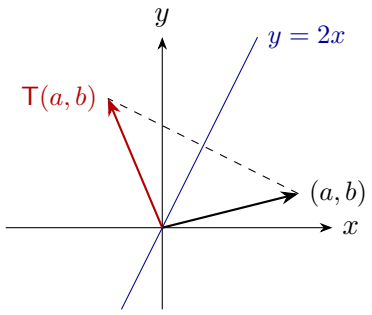


Figure. *Example 3: reflection about $y = 2x$.*

Corollary

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ and an ordered basis γ of $\mathbb{F}^n$, $[\mathsf{L}_A]_{\gamma} = Q^{-1}AQ$, where the j-th column of Q is the j-th vector of γ.

Proof. Theorem 2.23 with β standard: $[\mathsf{L}_A]_{\beta} = A$ and $Q = [\mathsf{I}]_{\gamma}^{\beta}$ has columns the vectors of γ. ■

Example 4

For $\gamma = \{(-1,0,0),(2,1,0),(1,1,1)\}$,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad [\mathsf{L}_A]_\gamma = Q^{-1}AQ = \begin{pmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{pmatrix}.$$

Definition (Similar matrices)

B is **similar** to A if $B = Q^{-1}AQ$ for some invertible Q; similarity is an equivalence relation. Theorem 2.23: matrices of one operator in two bases are similar.

2.6 Dual spaces *

Motivation. Linear functionals, the linear maps $V \to \mathbb{F}$, form the dual space V^*, of the same dimension as V. Coordinate functions give the dual basis; transposes of matrices are matrices of dual maps; and V is naturally isomorphic to V^{**}.

A **linear functional** on V is a linear map $V \to \mathbb{F}$.

Example 1

For continuous g on $[0,2\pi]$, $h(x) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$ is a linear functional; for $g(t) = \sin nt$ or $\cos nt$ it gives Fourier coefficients.

Example 2

$\operatorname{tr} : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is a linear functional.

Example 3

For an ordered basis $\beta = \{x_1,\dots,x_n\}$, the **$i$-th coordinate function** $f_i(x) = a_i$, where $[x]_\beta = (a_1,\dots,a_n)^t$, is a linear functional, and $f_i(x_j) = \delta_{ij}$.

Definition (Dual space)

$V^* = \mathcal{L}(V,\mathbb{F})$. If $\dim V < \infty$, $\dim V^* = \dim V \cdot \dim \mathbb{F} = \dim V$ (Corollary to Theorem 2.20), so $V \cong V^*$. The **double dual** is $V^{**} = (V^*)^*$.

Theorem 2.24

For an ordered basis $\beta = \{x_1,\dots,x_n\}$ with coordinate functions f_i, $\beta^* = \{f_1,\dots,f_n\}$ is an ordered basis of V^*, and $f = \sum_{i=1}^n f(x_i)f_i$ for every $f \in V^*$.

Proof. $\dim V^* = n$, so it suffices that β^* generates V^* (Corollary 2 of Theorem 1.10). For $g = \sum f(x_i)f_i$, $g(x_j) = \sum_i f(x_i)\delta_{ij} = f(x_j)$, so $g = f$ by the Corollary to Theorem 2.6. ∎

β^*, characterised by $f_i(x_j) = \delta_{ij}$, is the **dual basis** of β.

Example 4

For $\beta = \{(2,1),(3,1)\}$: solving $f_1(2,1)=1$, $f_1(3,1)=0$, $f_2(2,1)=0$, $f_2(3,1)=1$ gives $f_1(x,y) = -x+3y$ and $f_2(x,y) = x-2y$.

Theorem 2.25

Let V, W be finite-dimensional with ordered bases β, γ. For linear $\mathsf{T}: V \to W$, the map $\mathsf{T}^t: W^* \to V^*$, $\mathsf{T}^t(g) = g\mathsf{T}$, is linear and $[\mathsf{T}^t]_{\gamma^*}^{\beta^*} = ([\mathsf{T}]_\beta^\gamma)^t$.

Proof. $g\mathsf{T}$ is linear (Theorem 2.9), and $\mathsf{T}^t(ag+h) = (ag+h)\mathsf{T} = a\,g\mathsf{T} + h\mathsf{T}$. Let $\beta = \{x_s\}$, $\gamma = \{y_k\}$, $\beta^* = \{f_s\}$, $\gamma^* = \{g_j\}$, $A = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.24, $\mathsf{T}^t(g_j) = g_j\mathsf{T} = \sum_s (g_j\mathsf{T})(x_s) f_s$, so the (i,j) entry of $[\mathsf{T}^t]_{\gamma^*}^{\beta^*}$ is

$$(g_j\mathsf{T})(x_i) = g_j\Big(\sum_k A_{ki} y_k\Big) = A_{ji}.$$

■

Example 5

$\mathsf{T}(p) = (p(0), p(2))$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$, standard bases: $[\mathsf{T}]_\beta^\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 2 \end{smallmatrix}\right)$. Directly, $\mathsf{T}^t(g_1) = g_1\mathsf{T}$ sends $p \mapsto p(0)$, i.e. f_1, and $\mathsf{T}^t(g_2)$ sends $a+bx \mapsto a+2b$, i.e. f_1+2f_2. So $[\mathsf{T}^t] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right) = ([\mathsf{T}]_\beta^\gamma)^t$.

For $x \in V$ define $\hat{x} \in V^{**}$ by $\hat{x}(f) = f(x)$; $\hat{x}$ is linear in f.

Lemma

If $\dim V < \infty$, $x \in V$ and $\hat{x}(f) = 0$ for all $f \in V^*$, then $x = 0$.

Proof. If $x \neq 0$, extend $\{x\}$ to an ordered basis $\beta = \{x_1 = x, \ldots, x_n\}$; the dual basis gives f_1 with $f_1(x) = 1 \neq 0$. ■

Theorem 2.26

For finite-dimensional V, $\psi: V \to V^{**}$, $\psi(x) = \hat{x}$, is an isomorphism.

Proof. ψ is linear: $\widehat{cx+y}(f) = f(cx+y) = c\hat{x}(f) + \hat{y}(f)$. By the Lemma it is one-to-one, and $\dim V^{**} = \dim V$, so it is an isomorphism by Theorem 2.5. ■

(ψ uses no basis: it is a *natural* isomorphism.)

Corollary

If V is finite-dimensional, every ordered basis of V^* is the dual basis of some basis of V.

Proof. Let $\{f_1, \ldots, f_n\}$ be a basis of V^*, with dual basis $\{\hat{x}_1, \ldots, \hat{x}_n\}$ in V^{**} (Theorems 2.24, 2.26). Then $\delta_{ij} = \hat{x}_i(f_j) = f_j(x_i)$, so $\{f_j\}$ is the dual basis of $\{x_1, \ldots, x_n\}$, which is a basis because ψ is an isomorphism. ■

2.7 Homogeneous linear differential equations with constant coefficients *

Motivation. The solutions of $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y = 0$ form the null space of a differential operator $p(D)$. Linear algebra shows this space has dimension exactly n, and factoring the auxiliary polynomial $p(t)$ produces an explicit basis of exponentials.

A spring with constant k and mass m satisfies $y'' + \frac{k}{m}y = 0$.

Example 1

$y(t) = \sin\sqrt{k/m}\,t$ solves $y'' + \frac{k}{m}y = 0$, while $y(t) = t$ does not.

A **homogeneous linear differential equation with constant coefficients** is

$$a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \qquad a_i \in \mathbb{C},\ a_n \neq 0. \tag{2.1}$$

Dividing by a_n we assume $a_n = 1$. Solutions are sought among complex-valued functions of a real variable.

Definition (Derivative of complex functions)

$x = x_1 + ix_2 \in \mathcal{F}(\mathbb{R}, \mathbb{C})$ is **differentiable** if x_1, x_2 are, with $x' = x_1' + ix_2'$.

Example 2

$x(t) = \cos 2t + i \sin 2t$ has $x'(t) = -2\sin 2t + 2i\cos 2t$; and $x^2 = \cos 4t + i \sin 4t$.

Theorem 2.27

Every solution of (2.1) has derivatives of all orders.

Proof. If x is a solution, $x^{(n)} = -\sum_{i<n} a_i x^{(i)}$ is differentiable (each $x^{(i)}$, $i < n$, is). Inductively, if $x^{(k)}$ exists for all $k \leq m$ with $m \geq n$, then $x^{(m)} = -\sum_{i<n} a_i x^{(i+m-n)}$ is differentiable. ■

Example 3

For $y'' + 4y = 0$, $y'' = -4y$, so $y''' = -4y'$, and so on.

Definition (C^∞; differential operators)

C^∞ is the subspace of $\mathcal{F}(\mathbb{R}, \mathbb{C})$ of functions with derivatives of all orders. $D : C^\infty \to C^\infty$, $D(x) = x'$, is linear. For a complex polynomial $p(t) = \sum a_i t^i$ of positive degree, $p(D) = \sum a_i D^i$ is a **differential operator** of order $\deg p$. For (2.1) (with $a_n = 1$), $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ is the **auxiliary polynomial**, and (2.1) reads $p(D)(y) = 0$.

Theorem 2.28

The solutions of (2.1) are exactly $\mathrm{N}(p(D))$.

Proof. By Theorem 2.27 all solutions lie in C^∞, where the equation is $p(D)(y) = 0$. ■

Corollary

The solution set of (2.1) is a subspace of C^∞, the **solution space**.

Proof. Theorems 2.28 and 2.1. ■

Definition (Complex exponential)

For $c = a+ib$, $e^c = e^a(\cos b + i \sin b)$ (**Euler's formula**: $e^{ib} = \cos b + i \sin b$). Then $e^{c+d} = e^c e^d$ and $e^{-c} = 1/e^c$. $f(t) = e^{ct}$ is an **exponential function**.

Theorem 2.29

$\frac{d}{dt} e^{ct} = ce^{ct}$.

Proof [Supplement]. With $c = a + ib$, $e^{ct} = e^{at} \cos bt + ie^{at} \sin bt$, whose derivative is $e^{at}(a \cos bt - b \sin bt) + ie^{at}(a \sin bt + b \cos bt) = (a + ib)e^{at}(\cos bt + i \sin bt)$. ■

Theorem 2.30

The solution space of $y' + a_0 y = 0$ has dimension 1 and basis $\{e^{-a_0 t}\}$.

Proof. $e^{-a_0 t}$ is a solution by Theorem 2.29. If x is a solution, then $(e^{a_0 t}x)' = e^{a_0 t}(a_0 x + x') = 0$. A complex function with zero derivative is constant (apply the real fact to its real and imaginary parts). So $x = ke^{-a_0 t}$. ■

Corollary

For $c \in \mathbb{C}$, $\mathrm{N}(D - c\mathsf{I})$ has basis $\{e^{ct}\}$.

Proof. Theorem 2.30 with $a_0 = -c$. ■

Theorem 2.31

If c is a zero of the auxiliary polynomial $p(t)$, then e^{ct} is a solution.

Proof. Write $p(t) = q(t)(t-c)$. Polynomials in D commute, so $p(D) = q(D)(D - c\mathsf{I})$ and $p(D)(e^{ct}) = q(D)((D - c\mathsf{I})e^{ct}) = q(D)(0) = 0$ by the Corollary. ■

Example 4

$y'' - 3y' + 2y = 0$: $p(t) = (t - 1)(t - 2)$, so e^t and e^{2t} are solutions.

Theorem 2.32

For any differential operator $p(D)$ of order n, $\mathrm{N}(p(D))$ is an n-dimensional subspace of C^∞.

Lemma 1

$D - c\mathsf{I} : C^\infty \to C^\infty$ is onto for every $c \in \mathbb{C}$.

Proof. Let $v \in C^\infty$ and $w = ve^{-ct}$. Its real and imaginary parts have antiderivatives, giving $W \in C^\infty$ with $W' = w$. Put $u = We^{ct}$. Then $(D - c\mathsf{I})u = W'e^{ct} + cWe^{ct} - cWe^{ct} = v$. ■

Lemma 2

Let T, U be linear operators on V with U onto and $\mathrm{N}(\mathsf{T})$, $\mathrm{N}(\mathsf{U})$ finite-dimensional. Then $\mathrm{N}(\mathsf{TU})$ is finite-dimensional and $\dim \mathrm{N}(\mathsf{TU}) = \dim \mathrm{N}(\mathsf{T}) + \dim \mathrm{N}(\mathsf{U})$.

Proof. Let $\{u_1, \ldots, u_p\}$ be a basis of $\mathrm{N}(\mathsf{T})$ and $\{w_1, \ldots, w_q\}$ one of $\mathrm{N}(\mathsf{U})$. As U is onto, choose v_i with $\mathsf{U}(v_i) = u_i$. We show $\beta = \{v_1, \ldots, v_p, w_1, \ldots, w_q\}$ is a basis of $\mathrm{N}(\mathsf{TU})$ with $p + q$ distinct vectors.

Step 1 (inside). $\mathsf{TU}(v_i) = \mathsf{T}(u_i) = 0$ and $\mathsf{TU}(w_j) = \mathsf{T}(0) = 0$.

Step 2 (spans). If $\mathsf{TU}(v) = 0$, then $\mathsf{U}(v) = \sum a_i u_i = \mathsf{U}(\sum a_i v_i)$, so $v - \sum a_i v_i \in \mathrm{N}(\mathsf{U})$ equals $\sum b_j w_j$.

Step 3 (independent). If $\sum a_i v_i + \sum b_j w_j = 0$, applying U gives $\sum a_i u_i = 0$, so all $a_i = 0$, then all $b_j = 0$. (Distinctness follows similarly.) ■

Proof of Theorem 2.32. Induction on n. For $n = 1$, $p(D)$ is a nonzero multiple of $D - c\mathsf{I}$, and Theorem 2.30 applies. For $n > 1$, factor $p(t) = q(t)(t - c)$ over $\mathbb{C}$ with $\deg q = n - 1$, so $p(D) = q(D)(D - c\mathsf{I})$. By Lemma 1, $D - c\mathsf{I}$ is onto; by induction and Theorem 2.30, $\dim \mathrm{N}(q(D)) = n - 1$ and $\dim \mathrm{N}(D - c\mathsf{I}) = 1$. Lemma 2 gives $\dim \mathrm{N}(p(D)) = n$. ■

Corollary

The solution space of any nth-order equation (2.1) is an n-dimensional subspace of C^∞.

Proof. Theorems 2.28 and 2.32. ■

Theorem 2.33

For distinct $c_1, \ldots, c_n \in \mathbb{C}$, $\{e^{c_1 t}, \ldots, e^{c_n t}\}$ is linearly independent.

Proof. Induction on n; $n = 1$ is clear. If $\sum_i b_i e^{c_i t} = 0$, apply $D - c_n\mathsf{I}$: $\sum_{i<n} b_i(c_i - c_n)e^{c_i t} = 0$. By induction $b_i(c_i - c_n) = 0$, so $b_i = 0$ for $i < n$, and then $b_n e^{c_n t} = 0$ gives $b_n = 0$. ■

Corollary

If the auxiliary polynomial of an nth-order equation (2.1) has n distinct zeros $c_1, \ldots, c_n$, then $\{e^{c_1 t}, \ldots, e^{c_n t}\}$ is a basis of the solution space.

Proof. They are n independent solutions (Theorems 2.31, 2.33) in an n-dimensional space. ■

Example 5

$y'' + 5y' + 4y = 0$: $p(t) = (t + 1)(t + 4)$, so the solutions are $b_1 e^{-t} + b_2 e^{-4t}$.

Example 6

$y'' + 9y = 0$: $p(t) = (t - 3i)(t + 3i)$, basis $\{e^{3it}, e^{-3it}\}$. Since $\cos 3t = \frac{1}{2}(e^{3it} + e^{-3it})$ and $\sin 3t = \frac{1}{2i}(e^{3it} - e^{-3it})$, $\{\cos 3t, \sin 3t\}$ is also a basis.

Lemma

If the auxiliary polynomial is $(t-c)^n$, then $\beta = \{e^{ct}, te^{ct}, \ldots, t^{n-1}e^{ct}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). $(D - c\mathsf{I})(t^k e^{ct}) = kt^{k-1}e^{ct}$, so $(D - c\mathsf{I})^n(t^k e^{ct}) = 0$ for $k < n$.

Step 2 (independent). If $\sum_k b_k t^k e^{ct} = 0$, multiply by e^{-ct}: the polynomial $\sum b_k t^k$ vanishes for all real t, so all $b_k = 0$. Since the space has dimension n (Theorem 2.32), β is a basis. ■

Example 7

$y^{(4)} - 4y^{(3)} + 6y^{(2)} - 4y' + y = 0$: $p(t) = (t-1)^4$, so the solutions are $(b_1 + b_2 t + b_3 t^2 + b_4 t^3)e^t$.

Theorem 2.34

If the auxiliary polynomial is $(t - c_1)^{n_1} \cdots (t - c_k)^{n_k}$ with distinct c_i, then $\{e^{c_1 t}, te^{c_1 t}, \ldots, t^{n_1-1}e^{c_1 t}, \ldots, e^{c_k t}, \ldots, t^{n_k-1}e^{c_k t}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). Factors of $p(D)$ commute, so each $t^j e^{c_i t}$ $(j < n_i)$, killed by $(D - c_i\mathsf{I})^{n_i}$ (Lemma), is killed by $p(D)$.

Step 2 (independent) [Supplement]. There are $n = \sum n_i$ functions, so it suffices to show independence. Suppose $\sum_i q_i(t)e^{c_i t} = 0$ with polynomials $\deg q_i < n_i$; we show all $q_i = 0$ by induction on k. For $a \neq 0$, $(D - c\mathsf{I})(q(t)e^{(c+a)t}) = (q' + aq)e^{(c+a)t}$, and $q \mapsto q' + aq$ preserves degree, hence is injective on polynomials. Apply $(D - c_k\mathsf{I})^{n_k}$: the k-th term vanishes (Step 1), and the others become $\tilde{q}_i e^{c_i t}$ with $\deg \tilde{q}_i = \deg q_i$. By induction all $\tilde{q}_i = 0$, so $q_i = 0$ for $i < k$. Then $q_k e^{c_k t} = 0$, so $q_k = 0$. ■

Example 8

$y''' - 4y'' + 5y' - 2y = 0$: $p(t) = (t-1)^2(t-2)$, so the solutions are $b_1 e^t + b_2 te^t + b_3 e^{2t}$.